

**ANALYSIS ON MULTISPECIES FLOW OF GASES RELEASED BY A VEHICLE
AT IDLING CONDITION USING CFD**

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Abstract

Multiphase flow is a common phenomenon in many industrial processes, amongst them the oil and gas industry. Due to the complexity of multiphase flow, development of reliable analysis tool is difficult. The purpose of this project is to compare multiphase models available in the ANSYS software Fluent to Define absorption coefficient for flue gases. Gravity and operating conditions for buoyancy driven flows based on the kinetic theory of multicomponent gases, multi-fluid approaches are derived. The aim of the present work is to implement this model in the LB solver, extend it to take account multiple chemical reactions. The temperature is modeled through separate distribution function and the flow distribution function is assumed to be independent of temperature. The performance has been checked for a binary diffusion flow and a counter-current propane/air reacting flow. The obtained results show that this model able to deal with multi-species flows and solves the multi-component system of equations.

Keywords: LB solver, distribution function, ANSYS Fluent, binary diffusion flow, air reacting flow.

Introduction

The emission of harmful gases from vehicles has become a significant environmental concern in recent times. One of the major contributors to these emissions is the idling of vehicles. In this context, the present project aims to investigate the multispecies flow of gases released by a vehicle at idling conditions using Computational Fluid Dynamics (CFD) simulations. The analysis will focus on the study of the flow behavior of different gases, such as nitrogen oxides (NO_x), carbon monoxide (CO), and hydrocarbons (HC), released by the vehicle. The objective of the study is to gain insights into the impact of idling conditions on the concentration and dispersion of these gases in the surrounding atmosphere. CFD simulations will be performed to create a virtual model of the vehicle and its surrounding environment. The simulations will consider the impact of different parameters, such as temperature, humidity, wind speed, and direction, on the dispersion of gases. The results obtained from the simulations will be analyzed using different visualization techniques to understand the behavior of the gases under different conditions.

Statement of the Problem

The problem is to analyze the multispecies flow of gases released by a vehicle at idling condition using computational fluid dynamics (CFD). This involves understanding the complex flow behavior of the mixture of gases released by the vehicle and how it interacts with the surrounding environment. The objective is to obtain a comprehensive understanding of the flow characteristics and determine the concentrations and distributions of various pollutants emitted by the vehicle at idling conditions. This information will be useful in evaluating the impact of the vehicle emissions on the environment and developing effective strategies to mitigate their effects.

Objectives of the study

- To evaluate the flow behavior and characteristics of the multispecies gas mixture released

by the vehicle at idling conditions.

- To determine the concentration and distribution of various pollutants in the gas mixture, Including carbon monoxide, nitrogen oxides, and particulate matter.
- To investigate the effect of environmental conditions, such as wind speed and direction, on the dispersion of the gas mixture.
- To identify potential areas of high concentration and pollution hotspots around the vehicle.
- To compare the results of the CFD simulations with experimental data, where available, and validate the accuracy of the simulations.

Review of Literature

Yuzhuo [1] studied on the dispersion of carbon monoxide (CO) in vehicular exhaust plumes using two computational methods: Lattice Boltzmann Method and Large Eddy Simulation. The results show that both methods are effective in predicting the CO concentration and dispersion characteristics. **Chao [2]** presents a detailed multi-fluid lattice Boltzmann scheme for modeling reacting multi-species flows, which is capable of simulating complex chemical reactions with high accuracy and efficiency. The proposed scheme is validated through several benchmark tests and demonstrates superior performance compared to existing methods. **Yifan [3]** presented a field investigation to estimate the exposure of pedestrians and drivers to vehicle exhaust during idling and starting conditions. The investigation was conducted in a busy urban area, and measurements were taken at different distances from the vehicles. **Sawant [4]** proposed a consistent lattice Boltzmann model for simulating multicomponent mixtures that includes a new approach for handling interactions between different components. The proposed model is validated through several benchmark tests and is shown to be more accurate and efficient compared to existing methods. **Ning [5]** presented an experimental and numerical study on the dispersion of motor vehicle pollutants under idle conditions. The investigation was conducted using a full-scale vehicle and a wind tunnel, and the results show that the pollutant dispersion is highly dependent on wind speed and direction. The study also uses a computational fluid dynamics model to simulate the pollutant dispersion. **Chikatamarla [6]** presents a lattice Boltzmann method (LBM) for direct numerical simulation of turbulent flows. The proposed method uses a single-relaxation-time collision model and is capable of simulating turbulent flows with high accuracy and efficiency. The method is validated through several benchmark tests and is shown to be capable of capturing important features of turbulent flows, including the development of turbulence statistics and the energy spectrum. **Owen [7]** in his paper presents a high-order adaptive algorithm for simulating multispecies gaseous flows on mapped domains. The algorithm uses a combination of a high-order finite volume method and a high-order adaptive mesh refinement technique to efficiently capture complex flow features while minimizing computational costs.

Research Methodology

The finite element method (FEM) is a numerical technique for solving problems which are described by partial differential equations or can be formulated as functional minimization. A domain of interest is represented as an assembly of finite elements. Approximating functions in finite elements are determined in terms of nodal values of a physical field which is sought. A continuous physical problem is transformed into a discretized finite element problem with unknown nodal values. For a linear problem a system of linear algebraic equations should be solved. Values inside finite elements can be recovered using nodal values. Two features of the FEM are worth to be mentioned:

- 1) Piece-wise approximation of physical fields on finite elements provides good precision even with simple approximating functions (increasing the number of elements we can achieve any precision).
- 2) Locality of approximation leads to sparse equation systems for a discretized problem. This helps to solve problems with very large number of nodal unknowns.

Figures 1-4 show the design and specifications of the model on which the analysis is done.

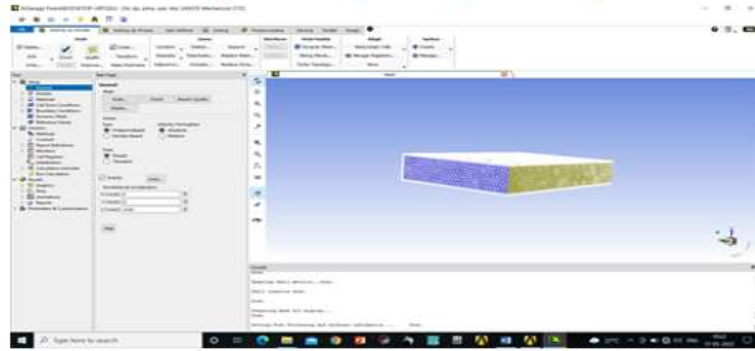


Figure 1: Ansys Fluent Workbench

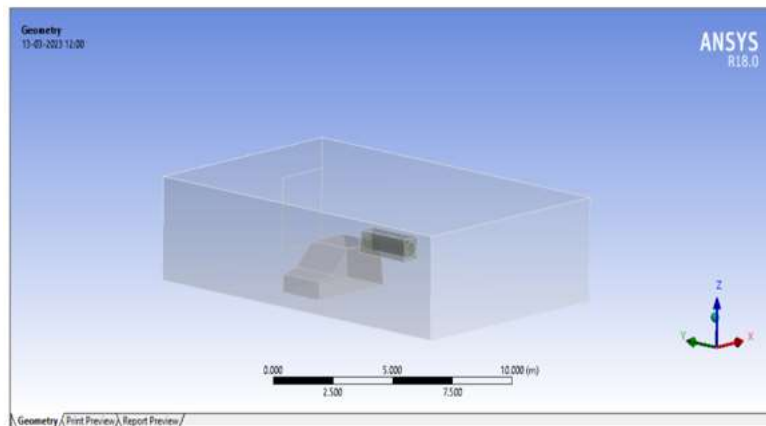


Figure 2: Design and Specifications

Specifications:

Name and Model: Honda Civic (2018)

Engine Capacity: 1.5 l

Displacement: 1498 CC

Torque: 219.64 n-m

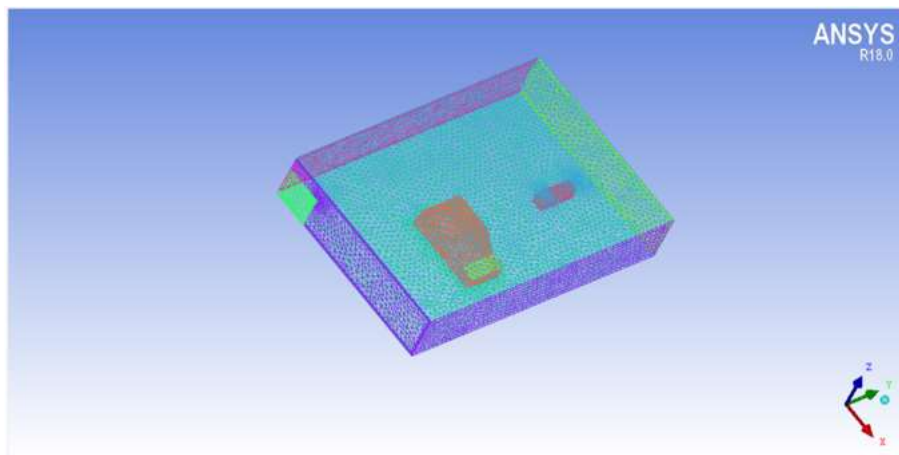


Figure 3: Meshed Design

Garage Dimensions:

Length: 15000 mm

Breadth: 10000 mm

Fan Dimensions:

Inner walls diameter: 550 mm

Jet Housing: 769.9 * 769.9 mm

Housing Length: 2500 mm

Car Dimensions:

Length : 4000 mm

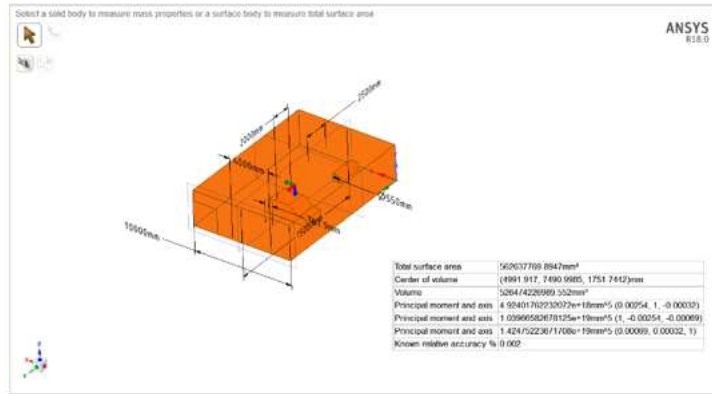


Figure 4: Model Dimensions

Results and Discussion

1. Concentration and distribution patterns of carbon monoxide (CO), nitrogen oxides (NO_x), and hydrocarbons (HCs) emitted from the exhaust of a vehicle at idling condition.
2. Identification of the impact of various factors, such as wind speed, vehicle speed, and atmospheric conditions, on the dispersion of these gases.
3. Validation of the developed model using experimental data, if available.
4. Recommendations for reducing the emission of harmful gases from vehicles and improving air quality in urban environments.

Figures 5-9 display the Post Processing results. Further assumptions are made based on these results.

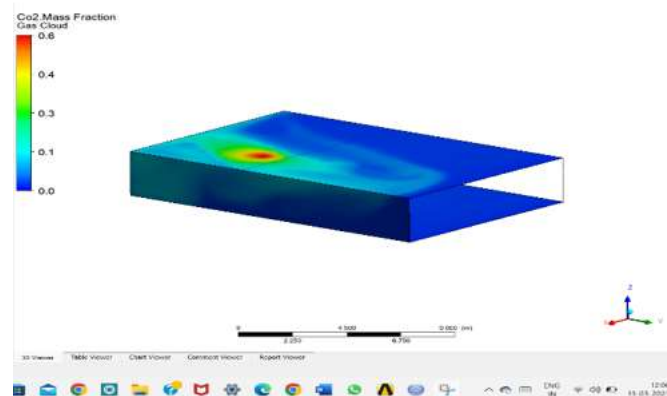


Figure 5: CO₂ Mass fraction

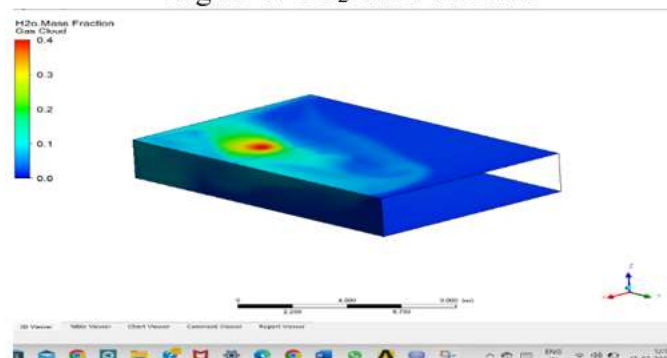


Figure 6: H₂O Mass fraction

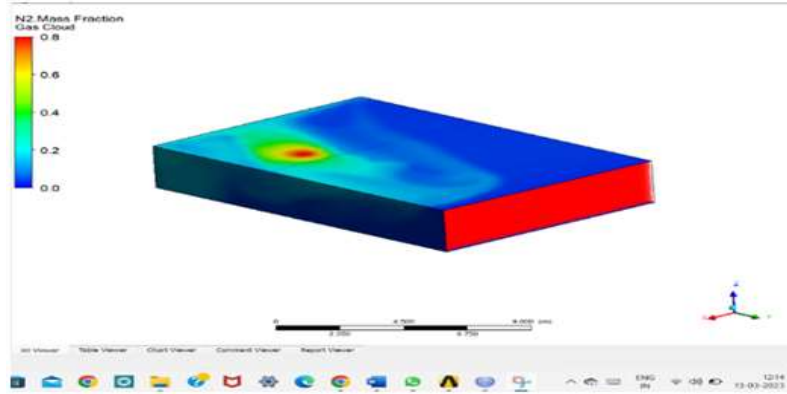


Figure 7: N₂Mass fraction

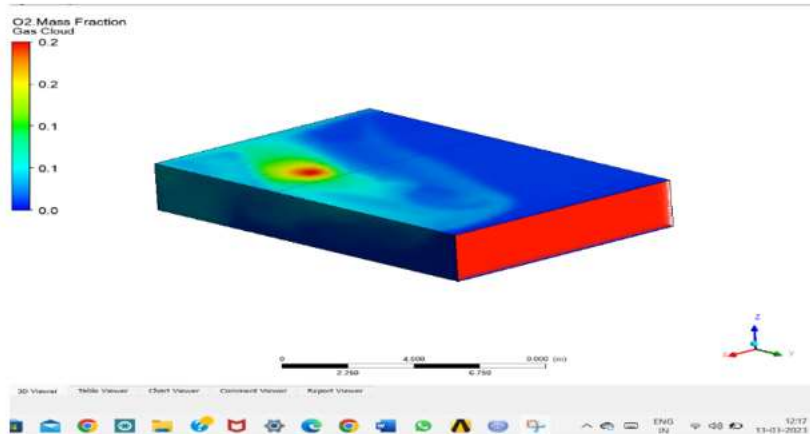


Figure 8 : O₂ Mass fraction

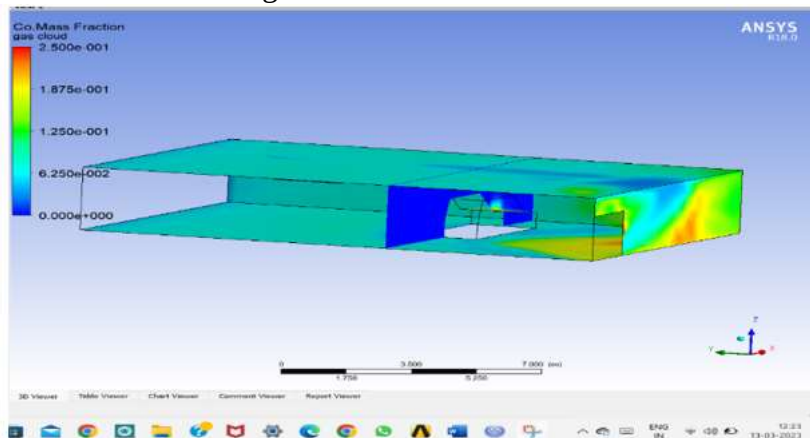


Figure 9: CO Mass fraction

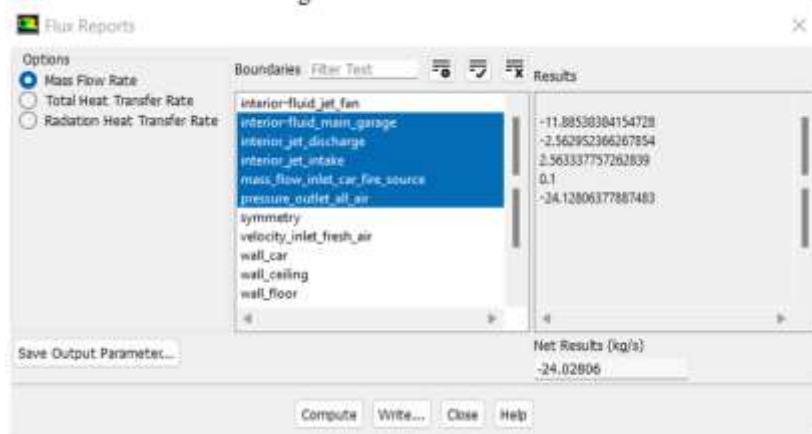


Figure 10: Flux Reports

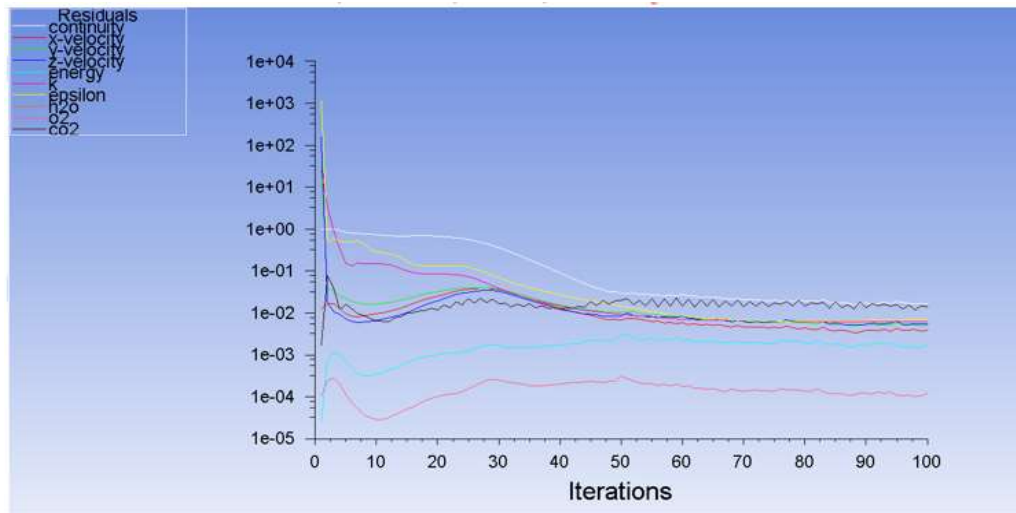


Figure 11: Scaled Residuals

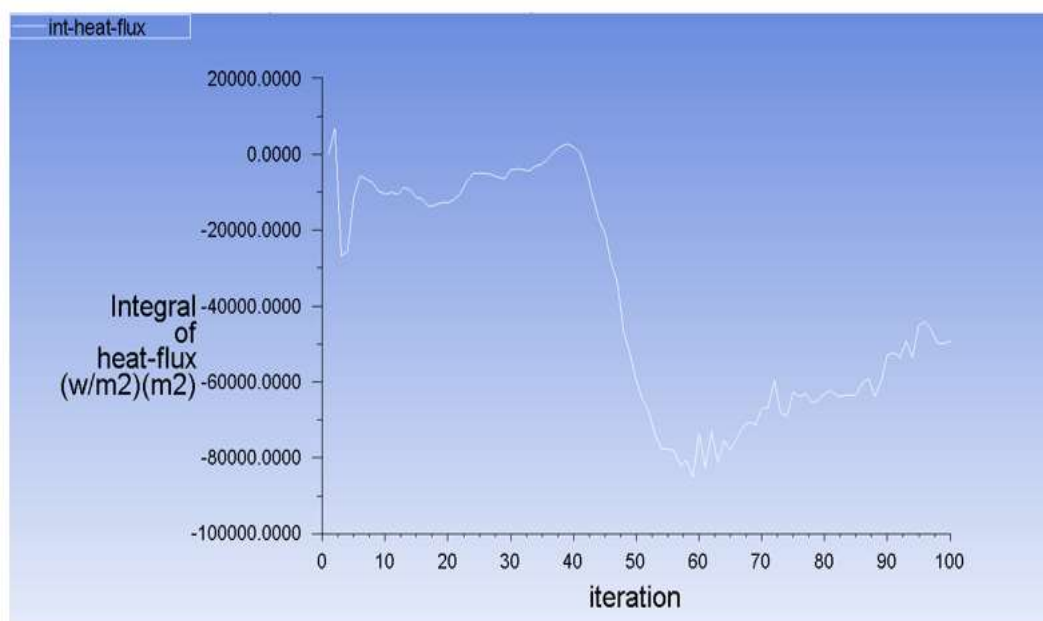


Figure 12: Mass flow rate

Results:

After Post Processing, These are the results are shown in Figures 10 to 12.

CO₂ Mass Fraction: 0.8714 m³

H₂O Mass Fraction: 0.662 m³

N₂ Mass Fraction: 279.84 m³

O₂ Mass Fraction: 83.68 m³

CO Mass Fraction: 0.211 m³

Conclusion:

The study provides valuable insights into the multispecies flow of gases, including carbon monoxide (CO), nitrogen oxides (NO_x), and hydrocarbons (HCs) emitted from a vehicle's exhaust at idling condition using Computational Fluid Dynamics (CFD) with Lattice Boltzmann Method (LBM). The analysis of the concentration and distribution patterns of these gases under various atmospheric conditions, wind speeds, and vehicle speeds suggests that they have a significant impact on their dispersion. The results of the study provide useful information that can help policymakers and regulators develop effective strategies for mitigating the environmental impact of vehicle emissions and improving air quality in urban areas. Further research is needed to validate the developed model and refine the findings of the study.

References

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